

# Paired-mutation obligation (task-text example / C- ) – documented, not implemented here

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Task line (tasks.md T054, verbatim, mutation clause): the sibling design brief for this class of task states the canonical mutation for ordering logic as "reverse the order key". Per `contracts/common-conventions.md` C-005: "Each tool also ships a paired §1.1 mutation of its own detection logic that flips a golden-bad to exit 0; the meta-test asserts the mutation is caught."

**This is T070's obligation, not T054's.** T054 (this RED test / this fixture set) exists specifically because `constitution/scripts/fastcycle/gates/gate_runner.sh` does not exist yet (see the main test's absence-proof Section A) — there is no ordering logic to mutate. Fabricating a premature mutation-test for code that does not exist would itself be a §11.4.6 violation (asserting a fact about an artefact's behaviour with nothing to observe).

**What T070's implementation must ship (named precisely, so this obligation is traceable and not lost per §11.4.197):** T070's implementation of

`gate_runner.sh` 's `--order history-cost sort` MUST have a paired `scripts/testing/meta_test_false_positive_proof.sh` mutation that **reverses the sort direction** of the `historical_fail_rate / mean_cost` ordering key — for example, changing the comparator from descending to ascending (or negating the computed ratio before sorting) — and re-running it against

`go_ordered_planted_fail_first/` (this directory's own golden fixture, whose `expected.json` says `execution_order_first_gate_id: "gate_c_planted_fail"` ) MUST observe the mutated tool wrongly report `gate_a_reliable` (the LOWEST-ranked gate) first instead — i.e. the golden test flips from PASS to FAIL under the reversed key. The meta-test then asserts that

mutation is caught (some OTHER check — golden-fixture regression, or the mutation-detection harness itself — FAILs when the reversal is applied), closing the loop C-005 requires.

This file is the durable record of that obligation so it is not silently dropped between T054 (RED, this task) and T070 (GREEN implementation).